

Emma M. Tinney, Ph.D.

Discovery Center for Brain Injury and Concussion Recovery
Spaulding Rehabilitation Hospital
Mass General Brigham
Harvard Medical School
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EDUCATION

Northeastern University, Boston MA

Doctor of Philosophy, April 2026

Advisor: Dr. Charles Hillman & Dr. Timothy Morris

Northeastern University, Boston MA

Master of Science in Psychology, March 2023

University of Delaware, Newark DE

Bachelor of Science, May 2021

Major: Neuroscience

Minor: Spanish

PROFESSIONAL EXPERIENCE

Spaulding Rehabilitation Hospital, Harvard Medical School, Boston MA

June 2026 – Present

Discovery Center for Brain Injury and Concussion Recovery

T32 Recovery and Restoration of CNS Health and Function Postdoctoral Research Fellow

Advisor: Dr. Simon Driver

RESEARCH

Department of Physical Therapy, Movement, & Rehabilitation Sciences, Northeastern University

Center for Cognitive and Brain Health

June 2025 – December 2025

PI: Timothy Morris, PhD

Project: **The Walking After Traumatic Brain Injury Study**, NCT06222502

Leading graduate student an NIH funded ongoing randomized clinical trial, which involves data collection, study management, participant recruitment, and data analysis to examine the effects of behavioral interventions on walking behaviors in adults with a history of TBI

Department of Physical Therapy, Movement, & Rehabilitation Sciences, Northeastern University

Center for Cognitive and Brain Health

June 2022 – October 2025

PI: Timothy Morris, PhD

Project: **The Exercise and Concussion Health Study (TECHS)**, NCT06494592

Spearheading an ongoing randomized clinical trial, which involves study design, data collection, study management, participant recruitment, and data analysis to examine the effects of exercise training on cognitive and brain health in patients recovering from concussion.

Department of Physical Therapy, Movement, & Rehabilitation Sciences, Northeastern University
Center for Cognitive and Brain Health June 2024 – August 2025

PI: Timothy Morris, PhD

Project: Functional MRI of Digital Trial Making Test

Developed novel interface for completing dTMT in the MRI scanner and investigate its construct validity by comparing performance and brain activation by comparing performance and brain activation between younger adults, older adults, and moderate/severe traumatic brain injury

Department of Psychology, Northeastern University

Center for Cognitive and Brain Health

2021-2024

PI: Charles Hillman, PhD & Arthur Kramer, PhD, 5R01AG053952

Project: Investigating Gains in Neurocognition in an Intervention Trial of Exercise (IGNITE)

Data collection and participant interaction in a NIH funded randomized clinical trial examining the effects of a year-long exercise training program on cognitive and brain health in older adults.

Department of Biomedical Engineering, University of Delaware

Mechanical Neuroimaging Lab

2019 – 2023

PI: Curtis L. Johnson, PhD, 5R01AG058853

Project: Mechanical Integrity of Memory Systems (MIMS) in Mild Cognitive Impairment

Performed data collection, analysis, and dissemination, including cognitive testing, neuroimaging, and fitness assessment, which assessed the mechanical integrity of memory systems in older adults with and without mild cognitive impairment, how the mechanical integrity of these systems is associated with memory performance, and whether integrity is modified by aerobic exercise training.

Department of Radiology, Brigham and Women's Hospital

2021 – 2022

PI: Samuel Patz, PhD, 1R01EB028664

Project: Magnetic Resonance Elastography of the Brain

Data collection and analysis of the implications of functional magnetic resonance elastography, aimed at developing ways to measure viscoelastic response functioning and tissue stiffness with high resolution

GRANTS & FUNDING

College of Science Equity Action Fund

2024

Fund to advance diversity, equity, inclusion, and justice for students within the College of Science at Northeastern University

Interdisciplinary Graduate Student Fellowship

2023

The fellowship requires work on an interdisciplinary research project with faculty members who have different areas of expertise and the research project has entail integrating information from different research domains for successful completion. Fellowship covers stipend and course fees.

HONORS & AWARDS

NIH T32 Fellowship Recovery and Restoration of CNS Health and Function

2026-2028

PI: Hochberg & Lin, T32NS100663, Mass General Brigham, Harvard Medical School

Women in Medicine and Science Featured Presenter

2026

Mass General Brigham

Basic Science of Physical Activity and Aging Biology World Congress Award American College of Sports Medicine	2023
Travel Award Society for Psychophysiological Research	2022
College of Science Travel Grant Northeastern University	2021-26
PhD Network Travel Grant Northeastern University	2021-26
Delaware Rehabilitation Institute Summer Scholars Award University of Delaware	2020

PEER REVIEWED PUBLICATIONS

(*student mentee, corresponding author, +invited)

Published:

11. **Tinney, E.M.**, Nwakamma, M, Espana-Irla, G., Kong, L., Chen, C., Hwang, J., O'Brien, A., Perko, M., Sodermann, L., Caefer, J., Manczurowsky, J., Hillman, C.H., Stillman, A.M., Morris, T.P. (2026) The feasibility and efficacy of a virtual, symptom-guided aerobic exercise intervention to improve cognition in mild traumatic brain injury: A single-blind pilot randomized control trial with an active comparator group. *Journal of Head Trauma Rehabilitation* DOI: 10.64898/2025.12.11.25342088v1
10. **Tinney, E.M.**, Nwakamma, M, Espana-Irla, G., Perko, M., Sodermann, R.L., Caefer, J., Manczurowsky, J., Hillman, C.H., Stillman, A.M., Morris, T.P. (2026) The Exercise and Concussion Health Study (TECHS): Pilot and Feasibility Protocol. 49: 101608. *Contemporary Clinical Trials Communications* DOI: 10.1016/j.conctc.2026.101608
9. **Tinney, E.M.**, Warren, A.E.L., Ai, M., Odom, H. *, Obrien, A. *, Sutton, B., Kang, C., Hang, H., Wan, L., Oberlin, L, Grove, G., Jakicic, J.M., Collins, A.M., Sewell, K.R., Burns, J., Vidoni, E., McAuley, E., Kramer, A.F., Erikson, K., Hillman, C.H. (2025). White Matter Microstructure Mediates the Association between Cardiorespiratory Fitness and Cognitive Performance in Older Adults. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 11:3, e70125. DOI: 10.1002/trc2.70125
8. España-Irla, G. M., **Tinney, E.M.**, Ai, M., Nwakamma, M., Morris, T.P. (2025). Functional connectivity Patterns following mild traumatic brain injury and the association with longitudinal cognitive function *Human Brain Mapping*, 46: e70237. DOI: 10.1002/hbm.70237
7. Ai, M., **Tinney, E.M.**, España-Irla, G. M., Hillman, C.H., Kramer, A.F., Morris, T.P. (2025). Brain resting-state functional connectivity mediates the age-associated decline in physical activity engagement. *Journals of Gerontology Medical Science*, 80:6, DOI: 10.1093/gerona/glaf075
6. **Tinney, E.M.**, Warren, A.E.L., Ai, M., Odom, H. *, Obrien, A. *, Morris, T.P., Sutton, B., Kang, C., Hang, H., Wan, L., Oberlin, L, Burns, J., Vidoni, E., McAuley, E., Kramer, A.F., Erikson, K., Hillman, C.H. (2024). Understanding Cognitive Aging through White Matter: A Fixel-Based Analysis. *Human Brain Mapping*, 45: e70121. DOI: 10.1002/hbm.70121

5. **Tinney, E.M.**, Sodemann, R.L.*, Espana-Irla, G., Perko, M.*, Hillman, C.H., Morris, T.P. (2024). Choroid Plexus Volume Only Increases Early after Traumatic Brain Injury and Only in Those with Positive Head Computed Tomography Scans. *Neurotrauma Reports*, 5:1, 1173-1185 DOI: 10.1089/neur.2024.0093
4. **Tinney, E.M.**, Ai, M., España-Irla, G. M., Hillman, C.H., Morris, T.P. (2024). Physical Activity and Frontoparietal Network Connectivity in Traumatic Brain Injury. *Brain & Behavior*, 14, e70022. DOI: 10.1002/brb3.70022
3. **Tinney, E.M.**, España-Irla, G. M., Warren, A.E.L., Whitehurst, L. N., Stillman, A.M., Hillman, C.H., Morris, T.P. (2024). Axonal injury to the anterior internal capsule is associated with sleep disturbances and memory decline following traumatic brain injury. *Ann Clin Transl Neurol*, 11: 2314-2326 DOI: 10.1101/ACN3.52145
2. Cline, T.L., Morfini, F., **Tinney, E.M.**, Makarewycz1, E., Lloyd, K., Olafsson, V., Bauer, C.C.C., Kramer, A.F., Raine, L.B., Gabard-Durnam, L.J., Whitfield-Gabrieli, S., Hillman, C.H. (2024). Resting-State Functional Connectivity Change in Frontoparietal and Default Mode Networks After Acute Exercise in Youth. *Brain Plasticity*, 9, 5-20, DOI: 10.3233/BPL-240003
1. **Tinney, E.M.**, Loui, P., Raine, L.B., Hiscox, L.V., Delgorio, P.L., Kramer, M.K., Schwarb, H., Martens, C.R., Kramer, A.K., Hillman, C.H., Johnson, C.L. (2023). Influence of mild cognitive impairment and body mass index on white matter integrity assessed by diffusion tensor imaging. *Psychophysiology*, 60, e14306. DOI: 10.1111/PSYP.14306

Submitted:

19. Morris, T.P., **Tinney, E.M.**, Toral, S., O'Brien, A., Gobena, E., Hackman, L., Nwakamma, M, Perko, M.L., Orchard, E., Odom, H., Chen, C., Hwang, J., Stillman, A.M., Kramer, A.F., Espana-Irla, G. Planning, Reminders and Micro-Incentives to Walk After Traumatic Brain Injury: A Pilot Randomized Control Trial. *medRxiv*
18. **Tinney, E.M.**, Nwakamma, M, Espana-Irla, G., Perko, M., Kong, L., Chen, C., Hwang, J., O'Brien, A., Caefer, J., Manczurowsky, J., Hillman, C.H., Stillman, A.M., Morris, T.P. Aerobic exercise improves executive function after traumatic brain injury via changes to the functional connectivity of the anterior cingulate cortex *medRxiv*
17. España-Irla, G. M., **Tinney, E.M.**, Perko, M., Nwakamma, M., Noble, S., Morris, T.P. Traumatic Brain Injury Associated with Altered Corpus Callosum Microstructure in Females: Exploring the Roles of Menopause Timing and Hormone Therapy in UK Biobank. *Women's Health*
16. Anto-Ocrah, M., Ugboji, A.Y.E., Fatima, F., **Tinney, E.M.**, Espana-Irla, G., Snedaker, K Overlapping Neurological Syndromes: Long COVID Morbidity in Females with Persistent Post-Concussion Symptoms. *Women's Health*
15. Nwakamma, M, **Tinney, E.M.**, Caefer, J., Espana-Irla, G., Hackman, L., Perko, M., Sodermann, L., Ande, K., Stillman, A.M., Morris, T.P. Feasibility of a modified exertional bike test to threshold virtual aerobic exercise intervention sessions in community-dwelling individuals with mild traumatic brain injury. *Restorative Neurology and Neuroscience*
14. Nwakamma, M, **Tinney, E.M.**, Ingemi, B., Hillman, C.H., Morris, T.P. Resting-State EEG Spectral Power Alterations Following Traumatic Brain Injury: A Comprehensive Systematic Review and Meta-Analysis of Clinical and Injury-Related Moderators. *Brain Injury*

13. España-Irla, G. M., Perko, M., **Tinney, E.M.**, Nwakamma, M., Morris, T.P. Sex and Gender Influences on Lifestyle Factors in Traumatic Brain Injury: A Narrative Review. *SportRxiv* DOI: 10.51224/SRXIV.684

12. **Tinney, E.M.**, Ande, K., Nwakamma, M., España-Irla, G. M., Morris, T.P. The Bidirectional Relationship Between Lifestyle Behaviors and Outcomes in TBI: A Narrative Review. *Brain Injury*, DOI: *In Prep*:

Utiz, B., **Tinney, E.M.**, Morris, T.P., Yarossi, M., Tunik, E., Technological Advances in Neuropsychological Assessment: A Systematic Review of Digital Trail Making Tests. (*in prep*)

Tinney, E.M., Utiz, B., Wong, E., Xu, E., Lynch, E., Hillman, C.H. Singh, T., Morris, T.P. Yarossi, M. Functional Activation during the Trail Making Test in Acquired Brain Injury (*in prep*)

Engima Sports Related Concussion Working Group. Leveraging Big Data to Answer Big Questions in Sport-Related Concussion and Repetitive Head Impact Research: Challenges and Opportunities to Harmonize Multimodal Data (*in prep*) *Sports Medicine*

Mendez-Pino, E., Thakur, B., Kumar, R.G., Dams-O'Connor, K., Ketchum, J., Rabinowitz, A., Sander, A.M., Venkatesan, U., Walker, W.C., O'Neil-Pirozzi, T.M., Wilmoth, K., **Tinney, E.M.**, Driver, S., Pinto, S.M., Association of Cardiovascular Disease and Vascular Risk Factors with Cognitive Function after Traumatic Brain Injury. (*in prep*)

ORAL CONFERENCE PRESENTATIONS

([†]peer reviewed published abstract, [^]symposium, *undergraduate student mentee

6. [^]**Tinney, E.M.**, Pinto, S.M, Driver, S., Lifestyle Interventions to Enhance Cardiovascular Health and Cognitive Recovery After Traumatic Brain Injury. American College of Rehabilitation Medicine Conference Symposium, Washington DC 2026

5. **Tinney, E.M.**, Burtrand, C, Jason, K., Warren, A.E.L., Morris, T.P., Sutton, B., Kang, C., Hang, H., Wan, L., Grove, G., Rogers, R.J., Jakicic, J. M., Marsland, A.L., Burns, J., Vidoni, E., Karikari, T., Morris, J., McAuley, E., Kramer, A.F., Erikson, K., Hillman, C.H. Effects of a 12-month Aerobic Exercise Intervention on White Matter Microstructure: Results from IGNITE. European Conference for Sports Science, Switzerland 2026

4. **Tinney, E.M.**, Warren, A.E.L., Ai, M., Odom, H. *, Obrien, A. *, Morris, T.P., Sutton, B., Kang, C., Hang, H., Wan, L., Oberlin, L, Burns, J., Vidoni, E., McAuley, E., Kramer, A.F., Erikson, K., Hillman, C.H. Cardiorespiratory Fitness, White Matter, and Cognitive Performance. European Conference for Sports Science, July 2025

3. [†]**Tinney, E.M.**, Nwakamma, M., España-Irla, G.M., Hackman, L., Perko, M., Chen, C., Hwang, J., Stillman, A., Morris, T.P.,. International Brain Injury Association, March 2025

2. **Tinney, E.**, Loui, P., Kramer, A., Hillman, C., Johnson, C., Influence of mild cognitive impairment and health markers on white matter integrity assessed by diffusion tensor imaging. New England American College of Sports Medicine Spring Conference, April 2023

1. [†]**Tinney, E.**, Morris, T., Hiscox, L., Delgorio, P., Martens, C., Kramer, A., Hillman, C., Johnson, C. The associations between cardiorespiratory fitness, hippocampal health, and cognition in mild cognitive impairment. American College of Sports Medicine Conference, May 2022

CONFERENCE POSTERS

30. **Tinney, E.M.**, Nwakamma, M., Espana-Irla, G., Perko, M., Stillman, A.M., Hillman, C.H., Morris, T.P. Aerobic Exercise Enhances Executive Function Recovery After Traumatic Brain Injury Through Anterior Cingulate Cortex Connectivity. Celebration of Women in Medicine & Science Symposium, September 2026

29. Morris, T.P. **Tinney, E.M.**, Nwakamma, M., Perko, M.L., Chen, C., Hwang, J., O'Brien, A., Hackman, L., Gobena, E., Orchard, E., Odom, H., Toral, S., Kramer, A. F., Stillman, A.M., Espana-Irla, G. The effect of planning, reminders and micro-incentives on physical activity participation in older adults after traumatic brain injury. European Conference for Sports Science, Switzerland 2026

28. **Tinney, E.M.**, Nwakamma, M., Espana-Irla, G., Perko, M., Stillman, A.M., Hillman, C.H., Morris, T.P. Aerobic exercise improves cognition after mild traumatic brain injury through changes to functional connectivity. Institute for Cognitive and Brain Health Conference, May 2026

27. **Tinney, E.M.**, Utiz, B., Wong, E.*, Xu, E.*, Lynch, E.*, Hillman, C.H. Singh, T., Morris, T.P. Yarossi, M. Functional MRI of the Trail Making Test in Traumatic Brain Injury. Neuro Boston, December 2025

26. Card, M. R., Drollette, E. S., Raine, L. B., **Tinney, E.M.**, Zimmerman, E., Schott, N., Hillman, C.H. The Association Between Early Motor Developmental Milestone Attainment and Conflict Monitoring in Preadolescent Children. European Conference for Sports Science, Italy 2025

25. [†]Oberlin, L., Collins, A.M., Vidoni, E.D., **Tinney, E.M.**, Ripperger, H., Wan, L., Raine, L., Molina-Hidalgo, C., Grove, G., Huang, H., Urrea, P.S., Sewell, K., Karikari, T., Morris, J., Kang, C., Marsland, A., Burns, J., McAuley, E., Kramer, A.F., Hillman, C.H., Erikson, K.I. Relating Physical Activity to Cognitive Function, Brain Morphology, and AD-related Blood Biomarkers: Preliminary Findings from the IGNITE study. Alzheimer's Association International Conference, July 2025

24. [†]Lynch, E.*, Frenzilli, I.*, Utiz, B., **Tinney, E.M.**, Singh, T., Morris, T.P. Yarossi, M. Comparative Analysis of Input Devices for Digitized Trail Making Task. Cognitive Neuroscience Society Annual Conference, April 2025

23. [†]Nwakamma, M., **Tinney, E.M.**, Caefer, J., Espana-Irla, G., Hackman, L., Perko, M., Sodermann, L., Ande, K., Stillman, A.M., Morris, T.P. The Boston Exercise Symptom Threshold Test for Mild Traumatic Brain Injury Rehabilitation. International Brain Injury Association, March 2025

22. [†]Espana-Irla, G., Perko, M., **Tinney, E.M.**, Sodermann, L., Ai, M., Hackman, L., Nwakamma, M., Morris, T.P., Sex and Age-Specific Factors in Mild Traumatic Brain Injury Recovery: The Role of Brain Network Modularity in Females. International Brain Injury Association, March 2025

21. **Tinney, E.M.**, Warren, A.E.L., Ai, M., Odom, H., Obrien, A., Morris, T.P., Sutton, B., Kang, C., Hang, H., Wan, L., Oberlin, L., Burns, J., Vidoni, E., McAuley, E., Kramer, A.F., Erikson, K., Hillman, C.H. Aging, White Matter, and Cognition. American Association for the Advancement of Science, February 2025

20. [†]**Tinney, E.M.**, Warren, A.E.L., Ai, M., Odom, H., Obrien, A., Morris, T.P., Sutton, B., Kang, C., Hang, H., Wan, L., Oberlin, L., Burns, J., Vidoni, E., McAuley, E., Kramer, A.F., Erikson, K., Hillman, C.H. Understanding Cognitive Aging through White Matter: A Fixel Based Analysis. Society for Psychophysiological Research, October 2024
19. [†]Nwakamma, M, **Tinney, E.M.**, Caefer, J., Espana-Irla, G., Hackman, L., Perko, M., Sodermann, L., Ande, K., Stillman, A.M., Morris, T.P. The Boston Exercise Symptom Threshold Test for Mild Traumatic Brain Injury Rehabilitation. American College of Rehabilitation Medicine, October 2024
18. [†]**Tinney, E.M.**, Ai, M., Espana-Irla, G., Hillman, C.H., Kramer, A.F., Morris, T.P. Physical activity engagement associated with dorsolateral prefrontal cortex connectivity in traumatic brain injury. American College of Sports Medicine Conference, May 2024
17. **Tinney, E.M.**, Nwakamma, M., Hackman, L., Espana-Irla, G., Hillman, C.H., Morris, T.P. The Exercise and Concussion Health Study: Feasibility and Preliminary Analyses. Traumatic Brain Injury Conference, May 2024
16. **Tinney, E.M.**, Nwakamma, M., Hackman, L., Espana-Irla, G., Morris, T.P. TBI Lab Research Highlights. Brain Injury Association of Massachusetts Conference, March 2024
15. [†]Ai, M., **Tinney, E.M.**, Hillman, C.H., Spreng, N., Kramer, A.F., Geddes, M. Age-effects on white matter topology in older adults at high risk of Alzheimer's disease. Geriatric Society Association Conference, November 2023
14. [†]**Tinney, E.M.**, Hillman, C. H., Morris, T.P. The Impact of Traumatic Brain Injury on Sleep, Brain, and Cognition in Older Adults. Society for Psychophysiology Research Conference, September 2023
13. [†]Espana-Irla, G., Ai, M., **Tinney, E.M.**, Morris, T.P. Functional connectivity patterns following traumatic brain injury. Society for Psychophysiology Research Conference, September 2023
12. [†]Cline, T.L., **Tinney, E.M.**, Morfini, F., Raine, L., Gabard-Durnam, L., Whitfield-Gabrieli, S., Kramer, A.F., Hillman, C.H. Acute effects of a single bout of exercise on functional brain networks in children. Society for Psychophysiology Research Conference, September 2023
11. [†]Ai, M., Colina, A.N., **Tinney, E.M.**, Warren, A.E.L., Hillman, C.H., Spreng, N., Kramer, A.F., Geddes, M. The association between APOE4, hypertension, and fixel-based white matter micro-structure in older adults at risk for Alzheimer's disease. Society for Psychophysiological Research, September 2023
10. [†]**Tinney, E.M.**, Warren, A.E.L., Kramer, A.F., Hillman, C.H., Johnson, C.L. Fixel-based dMRI analysis examining associations of aerobic fitness and white matter in aging and MCI. American College of Sports Medicine Conference, May 2023
9. [†]Cline, T.L., Watrous, J.N.H., Nwakamma, M., **Tinney, E.M.**, McDonald, K.M., Morfini, F., Raine, L., Gabard-Durnam, L., Kramer, A.F., Whitfield-Gabrieli, S., Hillman, C.H. Multivariate Pattern Analysis of Childhood Functional Brain Network Connectivity After Acute Moderate-to-Vigorous Physical Activity. American College of Sports Medicine Conference, May 2023
8. [†]Patz, S., Hoge, W.S., Deng, B., O'Donnell, L., Zhang, F., Tie, Y., **Tinney, E.**, Sinkus, R., Breedlove, K., High Heterogeneity of White Matter Observed with short TE Magnetic Resonance Elastography. ISMRM & SMRT Annual Meeting & Exhibition, May 2023

7. [†]**Tinney, E.**, Loui, P. C., Kramer, A., Hillman, C., Johnson, C. White matter integrity, body mass index, and mild cognitive impairment. Society for Psychophysiology Research Conference, September 2022
6. [†]**Tinney, E.**, Loui, P. C., Kramer, A., Hillman, C., Johnson, C. The relationship between cognition and white matter integrity in mild cognitive impairment using diffusion tensor tractography. Cognitive Neuroscience Society Conference, April 2022
5. [†]Hiscox, L., **Tinney, E.**, Delgorio, P., McGarry, M., Lanzi, A., Ellison, J., Cohen, M., Martens, C., and Johnson, C. Hippocampal viscoelasticity is associated with risk of mild cognitive impairment. ISMRM & SMRT Annual Meeting & Exhibition, May 15, 2021
4. **Tinney, E.**, Characteristics of Mild Cognitive Impairment and the Effects on Brain Health. University of Delaware Undergraduate Senior Thesis, May 4, 2021
3. **Tinney, E.**, Hiscox, L., Delgorio, P., Martens, C., and Johnson, C. Multimodal Characterization of Hippocampal Integrity in Mild Cognitive Impairment: Combining Volumetry, Diffusion, Perfusion, and Viscoelasticity from Magnetic Resonance Imaging. Society for Neuroscience Annual Conference, January 12, 2021
2. **Tinney, E.**, Hiscox, L., Delgorio, P., Martens, C., and Johnson, C. Multimodal Characterization of Hippocampal Integrity in Mild Cognitive Impairment: Combining Volumetry, Diffusion, Perfusion, and Viscoelasticity from Magnetic Resonance Imaging. Biomedical Engineering Society Annual Meeting, October 17, 2020
1. **Tinney, E.**, Hiscox, L., and Johnson, C. Characteristics of Mild Cognitive Impairment and the Effects on Brain Health assessed using Multiple Imaging Modalities. Summer Scholars Celebratory Virtual Symposium, August 13, 2020

MENTEE LOCAL PRESENTATIONS

9. Odom, H*, Orchard, E.*, Hackman, L., **Tinney, E.M**, Espana-Irla, G., Nwakamma, M., O'Neil-Pirozzi, T., Morris, T.P. Effects of Reminders and Micro-Incentives on Mental Health in Adults with TBI. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2025
8. Yedidia, N.*, Nwakamma, M., **Tinney, E.M**, Morris, T.P. Tribulations of Running Aerobic Exercise Interventions in Traumatic Brain Injury. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2025
7. Xu, E.,* **Tinney, E.M**, Uitz, B., Wong, E., Yarossi, M., Morris, T.P. Validity of the Digital Trial Making Test in Assessing Traumatic Brain Injury. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2025
6. Chen, C.C.*, **Tinney, E.M**, Hwang, J., Espana-Irla, G., Nwakamma, M., Stillman, A., Morris, T.P. Axonal Injury and Actigraphic Assessment of Sleep Disturbances Following Mild Traumatic Brain Injury. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2025
5. Kong, L. *, Espana-Irla, G., **Tinney, E.M**, Nwakamma, M., Stillman, A., Morris, T.P. Factors Influencing Exercise Intervention Compliance in Adults with Traumatic Brain Injuries. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2025

4. Hwang, J.*, **Tinney, E.M**, Chen, C.C., Espana-Irla, G., Nwakamma, M., Stillman, A., Morris, T.P. Rest Activity Rhythms and Cognition Following Traumatic Brain Injury. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2025

3. Sodemann, R.L.*, **Tinney, E.M**, Espana-Irla, G., Perko, M.* Morris, T.P. Choroid plexus enlargement is not associated with traumatic brain injury. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2024

2. Perko, M.*, Espana-Irla, G., Sodemann, R.L.*, **Tinney, E.M**, Morris, T.P. Sex differences in functional connectivity following traumatic brain injury. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2024

1. Ande, K. *, **Tinney, E.M**, Morris, T.P., Aerobic Exercise as a Tool for Traumatic Brain Injury Recovery. Research, Innovation, Scholarship and Entrepreneurship Expo, April 2024

INVITED TALKS

Lecture, The Delaney Senior Living Community 2026
Move More, Think Better: How Exercise Shapes the Brain

Lecture, Binge Thinking, Boston Lecture Series 2026
Move More, Think Better: How Exercise Shapes the Brain

Guest Speaker, Harvard Football Players Health Study Meeting 2026
The feasibility and efficacy of a virtual, symptom-guided aerobic exercise intervention to improve cognition in mild TBI

European Conference for Sports Science Brain Health Symposium 2025
Diffusing the Uncertainty of Diffusion Weighted Imaging

Society for Psychophysiological Research Symposium 2024
Faces of the Future

Guest Speaker, Neuroscience Honors Society 2024
Invited talk for undergraduate students about graduate school and neuroscience

Guest Speaker, Science Club 2024
Invited talk for 4th grade girls interested in learning more about science

MENTORSHIP AND TEACHING

- Trained and taught Northeastern University Undergraduate Research Assistants (URAs). Every semester in the Cognitive and Brain Health Lab (CBH) URAs are trained on all laboratory protocols to become proficient in data entry, administering neurocognitive, academic performance, and biospecimen testing, assist on fitness testing, lead intervention sessions, and assist in neuroimaging data collection.
 - n=53
- Mentored undergraduate students through individual research projects, including conference poster presentations and publications
 - n=13

TEACHING EXPERIENCE

Graduate Teaching Assistant:

Responsibilities:

- Grade quizzes, presentations, homework, lab reports, final exams
- Oversee discussion groups
- Administer, create, and instruct experimental lab sessions
- Provide feedback on lab report writing and research proposals
- Meet with students to discuss and improve lab report writing
- Deliver guest lectures

Courses:

2022 & 2023

- PSYC 4600: Laboratory in Research Design (*1 section, Psychology Department*)

Guest Lecturer:

HONR 3310: *Neuroscience & Brain Health*, Northeastern University

Topic: Brain Health and Exercise

RELEVANT COURSES

Grant Writing For Young Investigators

Northeastern University

2022

Seminar in Clinical Neuroscience

Northeastern University

2022

Physical Activity and Exercise: Prescription, Measurement, and Testing

Northeastern University

2022

Special Topics: EEG

Northeastern University

2023

EXTRACURRICULAR, SERVICE, & VOLUNTEER

I prioritize service efforts to make psychology, neuroscience, and academia a supportive and inclusive environment, as well as engage in efforts to promote inclusion and equity outside of academia.

ENIGMA Head Injury Working Group

2025-Current

Research Computing Director Job Search Committee

Northeastern University

2025

Heal Through Movement

Co-Founder

2025

Free fitness classes for individuals healing from trauma, intimate partner violence & brain injuries, with supportive workouts & community connections

Achilles Boston*Board Member & Volunteer*

2022-present

Assist and guide individuals with disabilities and visual impairments on weekly runs and races

Application Statement Feedback Program*Core Team*

2021-2025

Feedback reviewer and editor, ASFP is an outreach program providing graduate school statement feedback to individuals from underrepresented and marginalized communities applying to psychology PhD programs, helped recruit and coordinate 648 reviews from 184 statement editors for 272 applicants in 2021

Psychologists for Equity and Inclusion Association (PIE)*Northeastern University*

2022-2025

organizing our colloquium DEI speaker series, maintaining the DEI website, and promoting equity

Psychology Graduate Student Association (Psyc GSA)*Northeastern University*

2022-2025

Advocating for students, by students

NextGen Psych Scholars*Mentor*

2022-2025

Providing mentorship and guidance to undergraduate students in psychology related fields

Graduate Student Mentorship Program (GSMP)*Northeastern University*

2022-2025

Providing mentorship and guidance to undergraduate students in neuroscience

Northeastern University Community Liaisons for Education Using Stem (NUCLEUS)*Northeastern University*

2021-2025

Outreach science communication to youth in high school

Crisis Text Line*Bilingual Volunteer*

2022-2024

Handled live chats in English and Spanish with individuals in crisis that texted for assistance

STEM Field Trip

2023

Supervised field trip to Northeastern University and assisted in various STEM related activities

BOSTEM Expo

2023

Organized and led activities for children ages 5-18 interested in STEM research

Diversity and Inclusion Committee*University of Delaware*

2019 – 2021

Attended education workshops and educated peers on the importance of diversity, equity, and inclusion. Focused on enhancing an environment of the student body and facilitating events surrounding diversity and inclusion.

Student-Athletes against Social Injustice*University of Delaware*

2020 – 2021

Inspired by the Black Lives Matter movement, a small group of athletes formed a task force to meet with various influential people, i.e., Governor Carney, local police force, Players Coalition, and others. The

goal was to use resources to create an influential movement within our student-athlete community and impact those around us.

REVIEWER EXPERIENCE

Ad hoc reviewer: *Brain Structure and Function, Neurology*

LEADERSHIP

University of Delaware Division I Swimming and Diving Team

Scholarship Receiving Athlete & Team Captain

2017 – 2021

Developed leadership and team-building skills, comfortable performing under pressure, and a strong work ethic, displaying discipline and commitment through participation in rigorous, year-round training.

MEMBERSHIPS

- American College of Rehabilitation Medicine
- European College of Sports Science
- Society for Psychophysiological Research
- American College of Sports Medicine
- Cognitive Neuroscience Society
- Society for Neuroscience
- Biomedical Engineering Society

SKILLS

- Phlebotomy
- Matlab
- Freesurfer software
- FSL
- EEGLab
- HAPPE software
- MRTrix3
- MRI Level 1 Certified
- IBM SPSS Statistics
- Terminal
- R studio
- REDCAP
- VO₂ Max (treadmill, ergometer, portable)
- Actigraphy
- Biospecimen analysis
- Neuropsychology & Cognitive Testing
- MRI
- CITI training
- CPR certified
- BODPOD/PEAPOD/DXA
- BioRAFT
- Python

REFERENCES

Charles H. Hillman, Ph.D., Professor, Director of Institute for Cognitive and Brain Health
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